Improving Room Occupancy and Revenue Performance of a Family-Run Guest House

A Proposal report for the BDM capstone Project



Submitted by

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Declaration Statement

Iam working on a Project Title “Improving Room Occupancy and Revenue Performance of a Family-Run Guest House in Jodhpur”. I extend my appreciation to Marwar Guest House, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate:
Name: Maanvardhan Singh
Date: 13 October 2026

1. Executive Summary and Title

Title: ImprovingRoomOccupancyandRevenuePerformance of a Family-Run Guest House in Jodhpur

The project focuses on Marwar Guest House, a family-run guest house on the ground floor of a family home opposite Hotel Royal Inn on Paota A Road, near the New Bus Stand in Jodhpur. Owned by Mr. Mahipal Singh and operating since 2022, it offers seven rooms to travellers, and the business is B2C.

The major business issue is that the guest house has no clear view of how its occupancy and revenue change over the year. Every guest is a walk-in, the busiest periods are the festive and winter seasons, and bookings are noted only in small handwritten notebooks. As a result, rooms may stay empty in lean months, the tariffs of Rs. 1,200 for an AC room and Rs. 800 for a non- AC room may not suit every season, and the records cannot yet show which rooms and guests earn the most.

The issues will be addressed by collecting primary data from the guest house's own notebook records and analysing it with descriptive statistics, trend analysis and revenue analysis, using Excel and Python. The expected outcome is a set of practical recommendations that help the guest house fill more rooms in lean months, set tariffs on evidence and keep better records. The project cannot solve every problem of the business, but it will enable the guest house to identify its bottlenecks and tackle a few of them..

2. Organisation Background

Marwar Guest House is a family-runguesthouse in Jodhpur, Rajasthan, owned by Mr. Mahipal Singh and operating since 2022. It is located opposite Hotel Royal Inn on Paota A Road, near the New Bus Stand. The guest house occupies the ground floor of the family home, which the family renovated with an investment of about Rs. 20 lakh so that it could be used for renting out rooms. It has seven rooms, of which three are air-conditioned at Rs. 1,200 per night and four are non-air-conditioned at Rs. 800 per night, and it also provides Wi-Fi and parking. Three staff members work at the guest house, two for cleaning and housekeeping and one as the receptionist. Most guests are travellers who choose it because the bus stand and the railway station are close by, and all of them check in in person, since there is no online booking. The business is therefore B2C, and its busiest periods are the festive and winter seasons. Jodhpur, known as the Blue City, attracts many visitors to Mehrangarh Fort and other heritage sites.

3. Problem Statement

3.1Occupancypattern. The guest house does not know in which months and on which days its rooms remain empty, or how far the festive and winter peaks fill them.

3.2 Walk-in dependence. All guests arrive as walk-ins, so occupancy depends on footfall near the bus stand and railway station, and the guest house does not track who its guests are or how long they stay.

3.3 Tariff effect. Rooms are priced at Rs. 1,200 and Rs. 800, but there is no clear evidence of whether these rates suit peak and lean seasons or which room type earns more.

3.4 Informal records. Bookings are written in small handwritten notebooks, which makes it difficult to see patterns in occupancy, revenue and guest stays.

4. Background of the Problem

A guest house earns money only when its rooms are occupied, and a room left empty on a given night cannot be sold again for that night. If all seven rooms were occupied, the guest house would earn Rs. 6,800 a night, from three AC rooms at Rs. 1,200 and four non-AC rooms at Rs. 800, so every empty room matters. The problems listed above arise from a mix of internal and external causes.

Internal causes. Bookings are recorded informally in small handwritten notebooks, which makes it hard to total monthly revenue or compare one month with another. With only three staff members, of whom one is the receptionist, there may be little time to maintain detailed records. The guest house also takes no online bookings, so every guest has to walk in. **External causes.** Tourism in Jodhpur is seasonal, with the busiest periods in the festive and winter seasons and quieter months in between. Many hotels and guest houses near the bus stand and the railway station compete for the same travellers, and many travellers now compare options and book rooms online, which a walk-in-only guest house may miss.

These causes work together, because seasonal swings expose the lack of records and the dependence on walk-ins leaves little room to plan ahead. The project will therefore study how these problems appear in the guest house's own data and which of them can be reduced.

5. Problem Solving Approach

5.1 Methods and justification

The project will follow four steps: understand the business, collect the data, clean and analyse it, and draw recommendations. Two to three meetings with the owner, with field notes of each, will confirm the problem, because the owner knows the seasons, the guests and the earlier efforts better than any outside source. Occupancy trend analysis by month and weekday, with a comparison of the festive, winter and other months, will show when rooms remain empty. Revenue analysis by room type will show how much the AC and non-AC rooms earn. A comparison of tariff, occupancy and revenue per available room will show whether the current rates suit each season. Grouping guests by arrival day and length of stay will show how walk-in guests behave. Each method matches one problem statement.

5.2 Data collection and justification

Only primary data will be collected, directly from the guest house. The project will gather as many months of records as the notebooks hold, ideally 12 to 24 months, covering check-in and check- out dates, room number and type, tariff charged, number of guests, length of stay, discounts given and payment mode. Monthly operating expenses such as electricity, staff salaries, laundry and maintenance will come from the owner's bills. These variables match the problem statements: dates and room type for occupancy, arrival and stay details for walk-in guests, tariff and discounts for pricing, and the notebooks themselves for record quality. Handwritten entries will be typed into a spreadsheet with a metadata sheet that describes each column. Guest names, phone numbers and identity details will not be collected, which protects guest privacy. A signed letter from the owner, photographs of the property and a short video interaction with the owner will prove that the data is original.

5.3 Analysis tools and justification

Excel will be used for data entry, quick checks and pivot tables, because the owner can open and understand it easily. Python, with pandas and matplotlib, will be used for cleaning, grouping, descriptive statistics and charts, because it handles large records quickly and makes the work repeatable. Cleaning will remove duplicates, fix date and tariff errors, and flag missing values, and the cleaned data will be cross-checked against the owner's bills. Descriptive statistics such as the mean, median and range will summarise occupancy and tariffs, and correlation will test the link between tariff and occupancy. Every chart will have a title, labelled axes and a stated reason for the chosen chart type.

6. Expected Timeline

6.1 Work Breakdown Structure

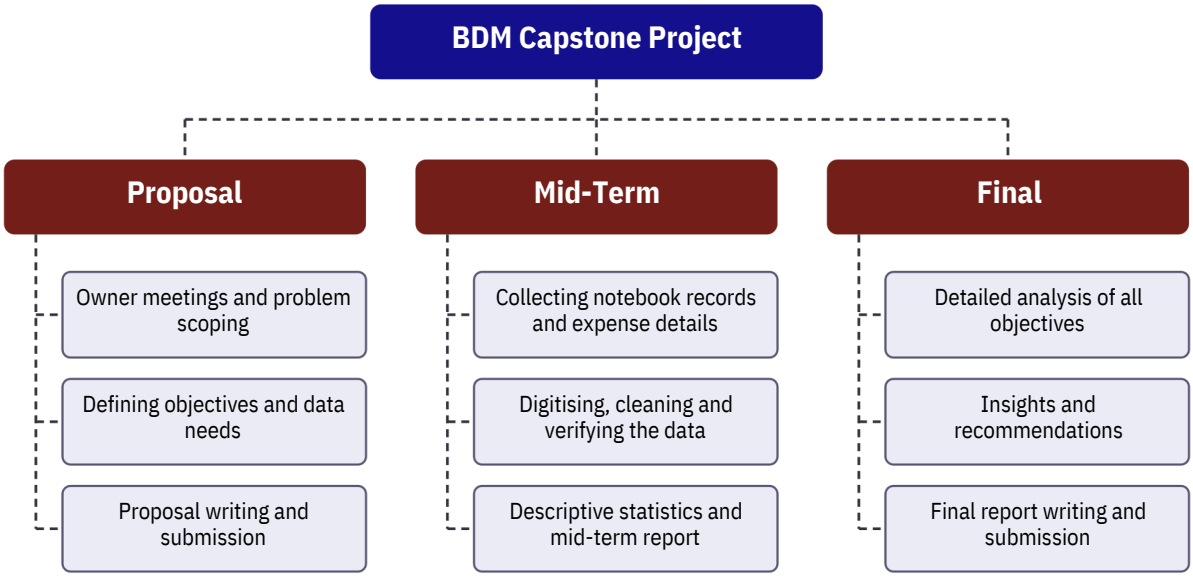


Figure 1: Work Breakdown Structure of the project

6.2 Schedule of tasks

The plan follows the submission windows of the course. The proposal will be submitted on 13 October 2026 (window of 11 to 13 October), the mid-term report in the window of 6 to 8 November 2026, and the final report in the window of 1 to 3 December 2026. Figure 2 shows when each task is planned, with weeks counted from Monday, 5 October 2026 (Week 1).

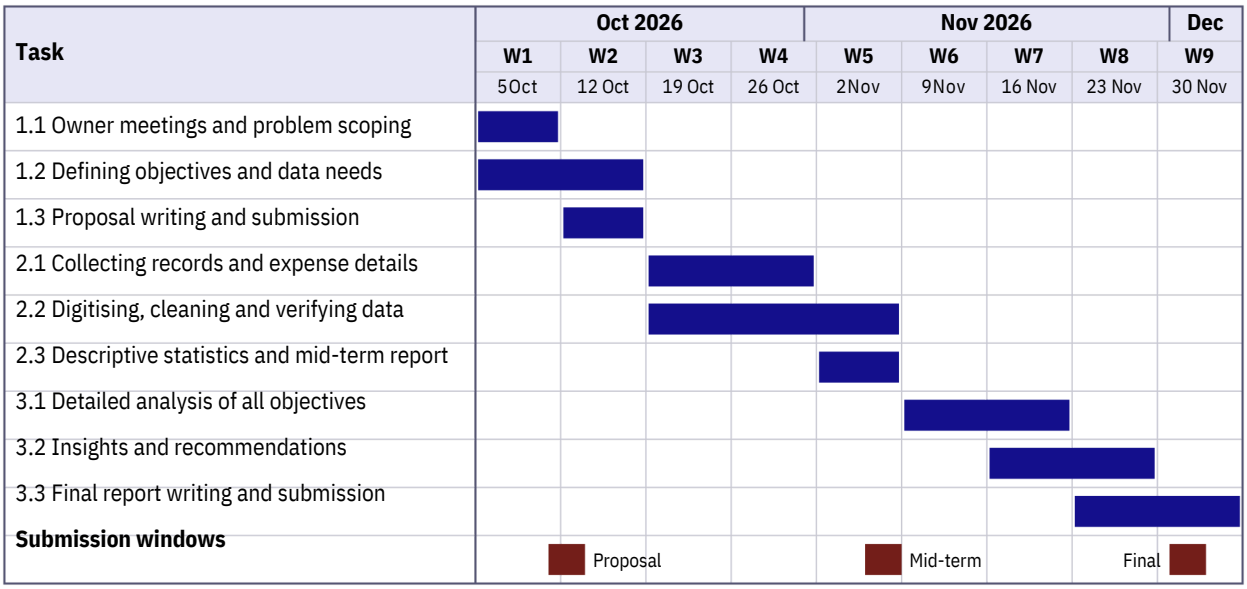


Figure 2: Gantt chart of the project tasks (maroon bars show the submission windows)

7. Expected Outcome

The project is expected to give the guest house the following outcomes.

7.1 Clear occupancy pattern. A month-wise and weekday-wise view of occupancy, with a comparison of the festive and winter peaks against the other months, which helps in planning staff, maintenance and promotions.

7.2 Evidence-based tariffs. A clear picture of how much the AC rooms at Rs. 1,200 and the non-AC rooms at Rs. 800 earn, and whether these rates suit each season, which supports seasonal rate setting.

7.3 Understanding of walk-in guests. Knowledge of when walk-in guests arrive, how long they stay and which periods are lean, which helps the owner judge whether simple ways of reaching guests beyond walk-ins are worth trying.

7.4 Simple digital records. A clean spreadsheet with metadata that the owner can keep updating in place of the informal notebooks, which makes future decisions easier.

Each outcome answers one problem statement listed in Section 3, and all insights will come from the analysis of the data and not from intuition. The project will therefore help the guest house identify its bottlenecks and tackle a few of them.